

Phase 6: User Interface Development

Project Title: Student Success & Career Guidance Portal

OVERVIEW:-

- Create an Apex controller (SkillController) that returns student skills.
- Build an LWC named skillTracker (HTML, JS, meta).
- Deploy Apex + LWC to your org.
- Create a Lightning App (Student Success) and edit the **Contact** record page to add: Related Lists, the skillTracker LWC, and a Report Chart.
- Test on a Student record.

6.1 Create Apex controller

1. In the org (Setup) — quick option:

- Setup → Quick Find → **Apex Classes** → **New** → write a code → **Save**.

Apex

```
public with sharing class SkillController {  
    @AuraEnabled(cacheable=true)  
    public static List<StudentSkillDTO> getSkillsForStudent(Id studentId){  
        if(studentId == null) return new List<StudentSkillDTO>();  
  
        List<Student__Skill__c> recs = [  
            SELECT Id, Proficiency__c,  
                Skill__c, Skill__r.Name, Skill__r.Skill_Level__c  
            FROM Student__Skill__c  
            WHERE Student__c = :studentId  
            ORDER BY Skill__r.Name  
        ];  
  
        List<StudentSkillDTO> out = new List<StudentSkillDTO>();
```

```

for(Student_Skill__c s : recs){
    String skillName = (s.Skill__r != null) ? s.Skill__r.Name : null;
    String skillLevel = (s.Skill__r != null) ? s.Skill__r.Skill_Level__c : null;
    out.add(new StudentSkillDTO(s.Skill__c, skillName, skillLevel,
s.Proficiency__c));
}
return out;
}

```

```

public class StudentSkillDTO {
    @AuraEnabled public Id skillId;
    @AuraEnabled public String skillName;
    @AuraEnabled public String skillLevel;
    @AuraEnabled public String proficiency;

    public StudentSkillDTO(Id skillId, String skillName, String skillLevel, String
proficiency) {
        this.skillId = skillId;
        this.skillName = skillName;
        this.skillLevel = skillLevel;
        this.proficiency = proficiency;
    }
}
}

```


Apex Classes

Apex Class

SkillController

Help for this Page ?

Apex Class Detail

Edit

Delete

Download

Security

Show Dependencies

Name	SkillController	Status	Active
Namespace Prefix		Code Coverage	0% (0/15)
Created By	Swayam Tamrakar , 9/21/2025, 9:59 AM	Last Modified By	Swayam Tamrakar , 9/21/2025, 9:59 AM

Class Body

Class Summary

Version Settings

Trace Flags

```

1 public with sharing class SkillController {
2     @AuraEnabled(cacheable=true)
3     public static List<StudentSkillDTO> getSkillsForStudent(Id studentId){
4         if(studentId == null) return new List<StudentSkillDTO>();
5
6         List<Student_Skill__c> recs = [
7             SELECT Id, Proficiency__c,
8                 Skill__c, Skill__rName, Skill__rSkill_Level__c
9             FROM Student_Skill__c
10            WHERE Student__c = :studentId
11            ORDER BY Skill__rName
12        ];
13
14        List<StudentSkillDTO> out = new List<StudentSkillDTO>();
15        for(Student_Skill__c s : recs){

```

Notes:

- This returns a small DTO list ready for LWC.
- cacheable=true lets the wire adapter cache results for performance.

6.2 Create the LWC (skillTracker) in VS Code

1. In VS Code

- Command palette → **SFDX: Create Project**.
- Connect to org: **SFDX: Authorize an Org** → choose your Developer Org.
- In the force-app/main/default/lwc folder → right-click → **SFDX: Create Lightning Web Component** → name it skillTracker.

You'll get three files. Replace them with the code below.

- **skillTracker.html**

```
<template>
```

```
<lightning-card title="Skills" icon-name="standard:skills">
```

```
<template if:true={skills.data}>
```

```
<div class="slds-p-around_small">
```

```

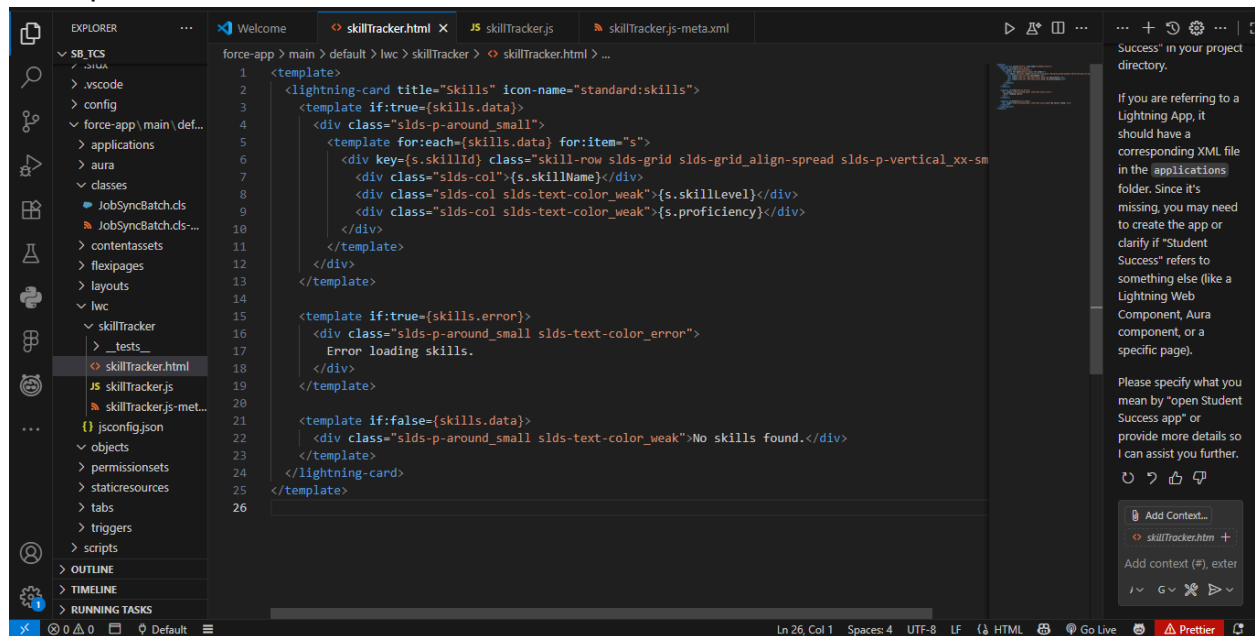
<template for:each={skills.data} for:item="s">
  <div key={s.skillId} class="skill-row slds-grid slds-grid_align-spread slds-p-vertical_xx-small">
    <div class="slds-col">{s.skillName}</div>
    <div class="slds-col slds-text-color_weak">{s.skillLevel}</div>
    <div class="slds-col slds-text-color_weak">{s.proficiency}</div>
  </div>
</template>
</div>
</template>

<template if:true={skills.error}>
  <div class="slds-p-around_small slds-text-color_error">
    Error loading skills.
  </div>
</template>

<template if:false={skills.data}>
  <div class="slds-p-around_small slds-text-color_weak">No skills found.</div>
</template>
</lightning-card>

```

</template>



- **skillTracker.js**

```
import { LightningElement, api, wire } from 'lwc';
```

```
import getSkillsForStudent from '@salesforce/apex/SkillController.getSkillsForStudent';
```

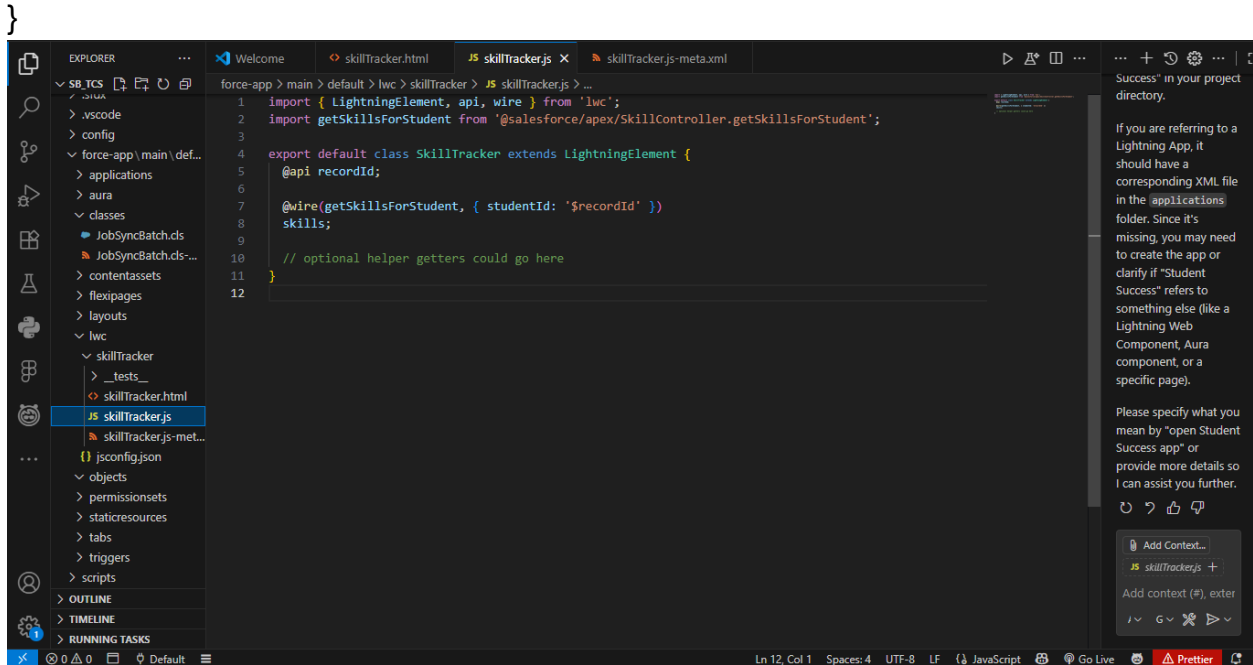
```
export default class SkillTracker extends LightningElement {
```

```
  @api recordId;
```

```
  @wire(getSkillsForStudent, { studentId: '$recordId' })
```

```
  skills;
```

```
  // optional helper getters could go here
```



- **skillTracker.js-meta.xml**

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<LightningComponentBundle xmlns="http://soap.sforce.com/2006/04/metadata">
```

```
  <apiVersion>59.0</apiVersion>
```

```
  <isExposed>true</isExposed>
```

```
  <targets>
```

```
    <target>lightning__RecordPage</target>
```

```
    <target>lightning__AppPage</target>
```

```
  </targets>
```

```
  <targetConfigs>
```

```
    <targetConfig targets="lightning__RecordPage">
```

```
      <objects>
```

```
        <object>Contact</object>
```

```
      </objects>
```

```
    </targetConfig>
```

</targetConfigs>

</LightningComponentBundle>

2. Deploy to Org

- In VS Code explorer → right-click component folder → **SFDX: Deploy Source to Org**.
- Or run CLI: `sfdx force:source:deploy -p force-app/main/default/lwc/skillTracker`.

6.3 Add the LWC & Components to Contact Record Page

1. Create Lightning App

- Setup → **App Manager** → **New Lightning App**
 - App Name: Student Success → Next → choose branding (color/logo) → Next.
 - Add Navigation Items: **Contacts, Internships** (your custom object), **Placements, Job_Posts, Reports, Dashboards, Student Skills, Skills, e.t.c** → Finish.

App Settings

- App Details & Branding
- App Options
- Utility Items (Desktop Only)
- Navigation Items
- User Profiles

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

* App Name ⓘ
Student Success

* Developer Name ⓘ
Student_Success

Description ⓘ
An app to track student performance, internships, placements, and provide AI-

App Branding

Image ⓘ


Primary Color Hex Value ⓘ
 #FAFE01

Clear

Org Theme Options
☐ Use the app's image and color instead of the org's custom theme

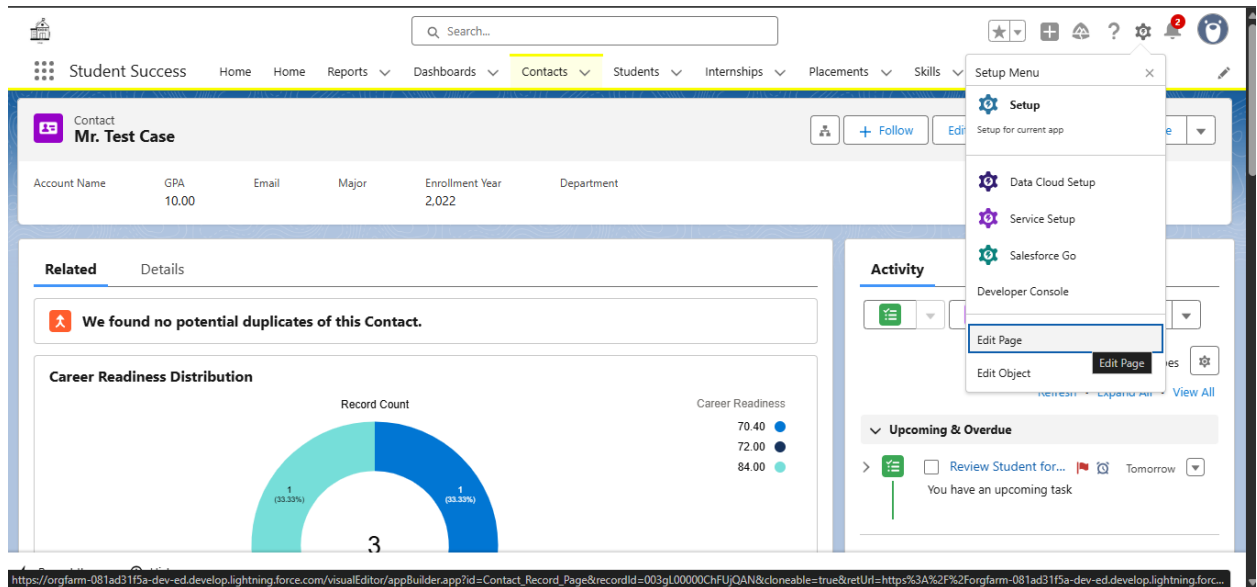
App Launcher Preview



Student Success
An app to track student performance, internships, plac...

2. Edit Contact Record Page

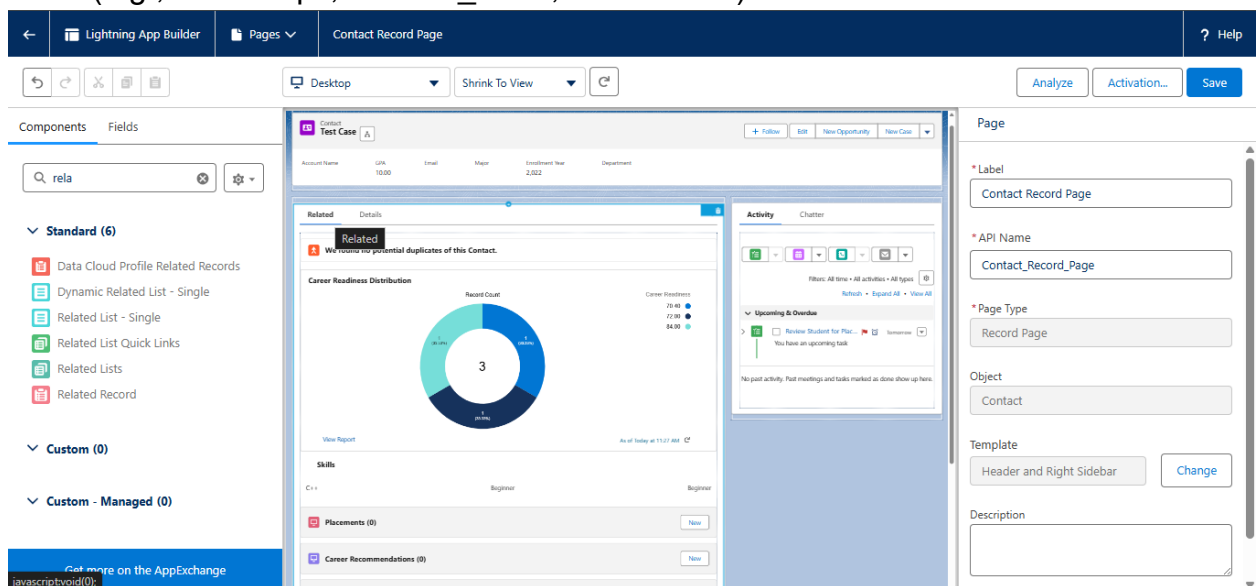
- App Launcher → open **Contacts** → open any Contact record → click the gear icon (⚙️) top-right → **Edit Page**. This opens **Lightning App Builder**.



- You're now editing the Contact Record Page. Do these:

Add Related Lists

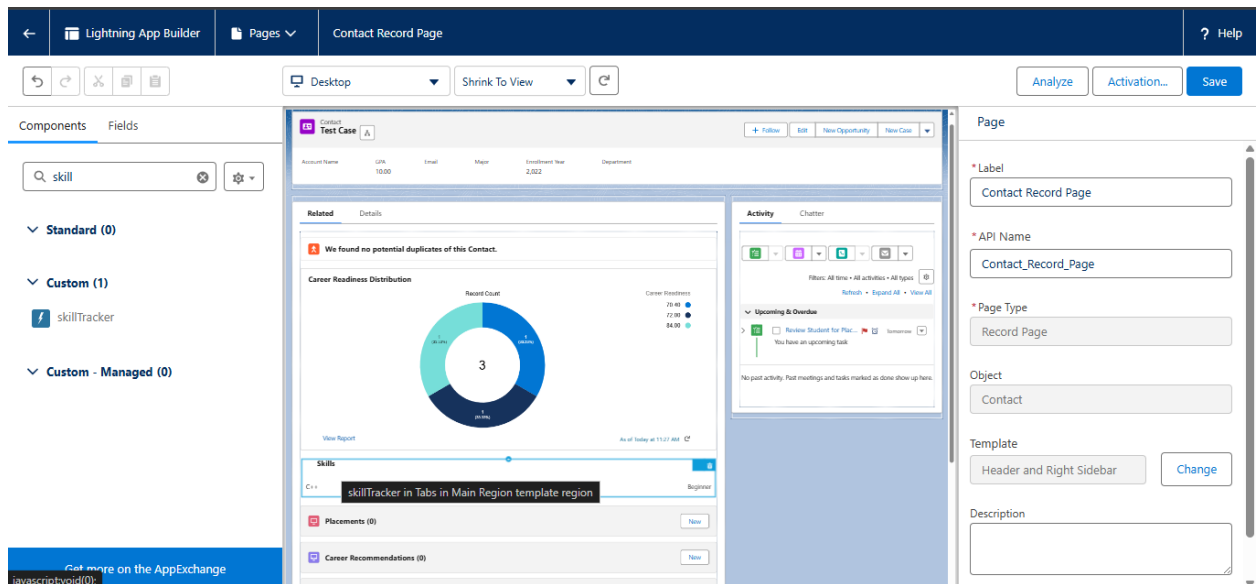
- From left components panel → drag **Related Lists** (or **Related List - Single**) onto the page where you want them.
- Click the related list component → in the right pane choose which related list to show (e.g., Internships, Student_Skills, Placements).



- Save & Activate (we'll set for the Student Success app or org default).

Add the skillTracker LWC

- In left panel → find **Custom** section → your skillTracker component → drag it onto the page (a column/region).



Add a Report Chart (Career Readiness distribution)

- Create a report.
 - Create a Custom Report Type for Student Skills
 - Go to **Setup**.
 - Quick Find → **Report Types** → click **New Custom Report Type**.
 - Configure:
 - ✓ **Primary Object**: Contacts
 - ✓ **Report Type Label**: Contacts with Student Skills
 - ✓ **Description**: Report showing Contacts and their related Student Skills.
 - ✓ **Category**: Choose Other Reports
 - ✓ **Deployment Status**: Deployed
 - Click **Next**.
 - Related Object: Student_Skill__c.
 - Choose: "Each A record must have at least one related B record."
 - **Save**
 - Create a Report from this New Report Type
 - App Launcher → **Reports** → **New Report**.
 - In Report Type selector, search for **Contacts with Student Skills**.

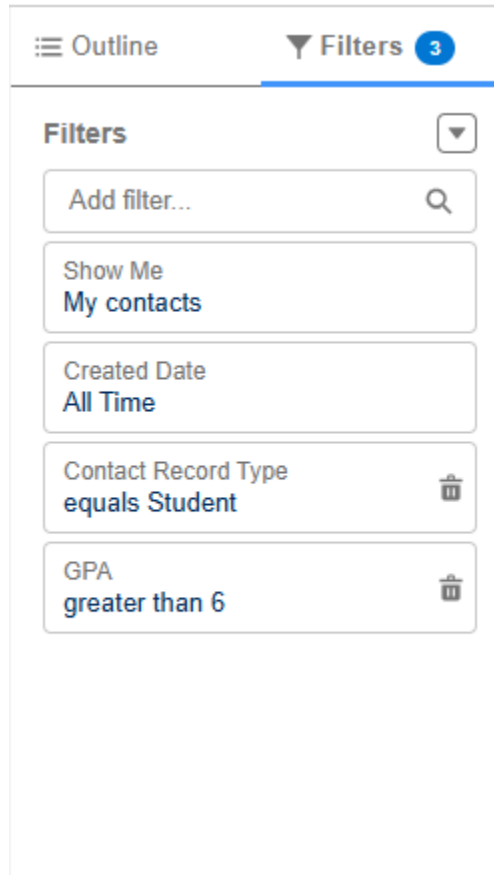
- Add Fields:
 - ✓ Name
 - ✓ Career_Readiness__c
 - ✓ Skill__r.Name
 - ✓ Proficiency__c

- Buckets:
 - ✓ In the left **Fields** pane, find the Career_Readiness__c field.
 - ✓ Click the chevron (▾) next to that field and choose **Bucket this field** (or click **Add | Bucket Field** at top of Fields pane).
 - ✓ Give the bucket a name: **Readiness Bucket**.
 - ✓ Choose **Bucket By: Numeric Range** (or the UI will allow you to create ranges).
 - ✓ Create ranges, for example:
 - Low : 0 to 30
 - Medium : 31 to 60
 - High : 61 to 100

- Group rows:
 - ✓ Drag the **Readiness Bucket** from **Fields** into the **Group Rows** area on the left (where it says *Add group...*).
 - ✓ You should now see **GROUP ROWS** populated — the report preview updates.

- Add Filter:
 - ✓ On the right side, open the **Filters** pane.
 - ✓ By default you may see a Date filter (e.g., Show Me / Created Date). Remove or set it to *All Time*.
 - ✓ Click **Add Filter**.
 - Field: type **Record Type** → choose **Record Type** (Contact.Record Type).
 - Operator: **equals**.
 - Value: choose **Student**

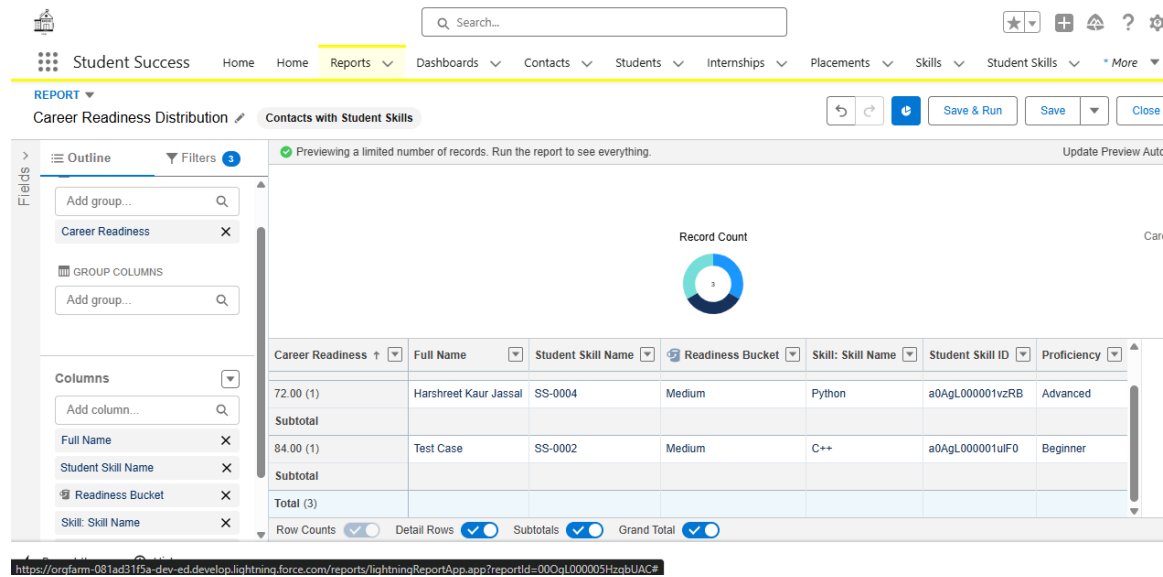
- Click **Apply**.



The screenshot shows a software interface with a top navigation bar containing 'Outline' and 'Filters 3'. The 'Filters' panel is open, displaying a list of filters. At the top is a search bar labeled 'Add filter...'. Below it are five filter items, each with a trash icon on the right: 'Show Me My contacts', 'Created Date All Time', 'Contact Record Type equals Student', and 'GPA greater than 6'. The fifth item is partially visible.

- ✓ Optional: add any other filters (Department, GPA > 6.0, etc).
- Add Chart:
 - ✓ Once you have a group, **Add Chart** becomes active (top bar). Click **Add Chart**.
 - ✓ In the chart configuration panel:
 - **Chart Type**: choose **Donut** (good for distribution) or **Bar**.
 - **Grouping**: should already be set to the group you made
 - **Values**: default is **Record Count**
 - Optional: show percentage or values in chart options.

- Click Done

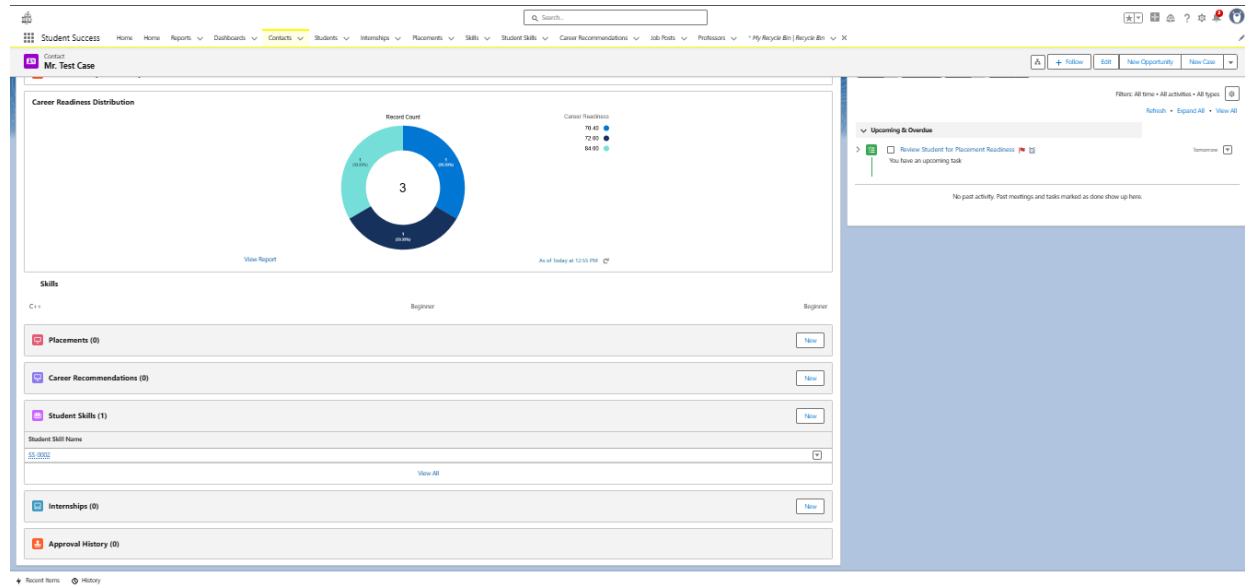


- Save & Run the report.
 - Click **Save & Run**.
 - Name the report: **Career Readiness Distribution**.
 - Choose a folder (e.g., My Personal Reports or a team folder).
 - Click **Save** → then **Run**.

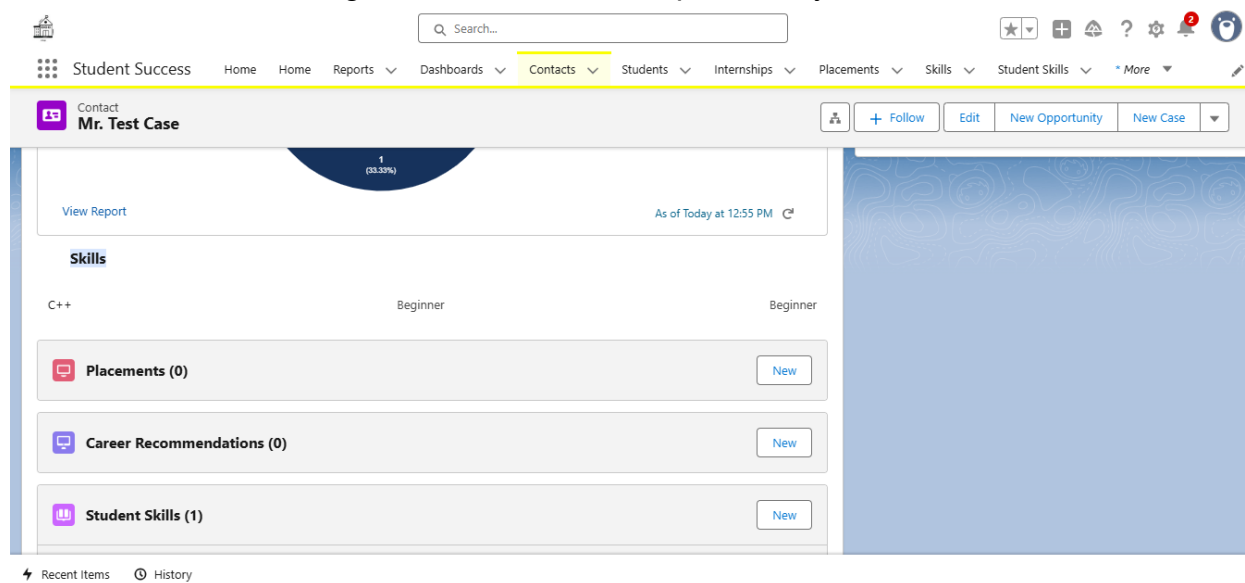
6.4 Test everything in Org

1. Create a **Contact** with Record Type Student. Add GPA and other fields.
2. Create a **Skill__c** record and a **Student_Skill__c** linking the Student.
3. Open the Student Contact record. The page should show:

- Related lists (Internships, Skills).



- The **Skills** LWC showing skill name, skill level, proficiency.



- The **Report Chart** with readiness distribution.

